

The Linear Algebra of Probability and Information

There are enough dot-product-like operations in probability that some linear algebra construction and notation can be warranted. The usefulness of this notation is still ambiguous.

Let's first represent a distribution as a vector $p_x = \begin{bmatrix} p(x=x_1) \\ p(x=x_2) \\ \vdots \\ p(x=x_n) \end{bmatrix}$.

If we have the two distributions/vectors p_x and p_y , what kind of relationship can we find between them? Bayes' rule gives us $p(x|y)p(y) = p(y|x)p(x) = p(x,y)$. Note that both $p(x,y)$ and $p(x|y)$ are functions of two variables, so it'd be appropriate to extend them as matrices.

For conditional distributions, we will use the notation. The joint will be similar.

$$C_{(x|y)} = \begin{bmatrix} p(x_1|y_1) & p(x_1|y_2) & \dots & p(x_1|y_n) \\ p(x_2|y_1) & p(x_2|y_2) & & \vdots \\ \vdots & & & \vdots \\ p(x_n|y_1) & \dots & \dots & p(x_n|y_n) \end{bmatrix} \quad \text{below} \quad J_{(x,y)} = \begin{bmatrix} p(x_1,y_1) & p(x_1,y_2) & \dots & p(x_1,y_n) \\ p(x_2,y_1) & p(x_2,y_2) & & \vdots \\ \vdots & & & \vdots \\ p(x_n,y_1) & \dots & \dots & p(x_n,y_n) \end{bmatrix}$$

The first parameter in the subscript varies along the vertical direction, and the second parameter varies along the horizontal direction. All of these matrices can be square with zero-padding.

Since $p(x,y) = p(y,x)$, we can verify that $J_{(x,y)}^T = J_{(y,x)}$ ①

Given what we know about Bayes' Rule, we can write $J_{(x,y)} = C_{(x|y)} D_y$ ②

Where $D_y = \begin{bmatrix} p(y_1) & & & \\ & p(y_2) & & \\ & & \ddots & \\ & & & p(y_n) \end{bmatrix}$.

If you add all the elements along a row of $J_{(x,y)}$, you will get an element of p_x . Let's call a vector of all ones $w = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$ ("one" sounds like "wun", hence the w). Now we can write $p_x = J_{(x,y)} w$ ③

Using ② and ③ $p_x = J_{(x,y)} w$
 $p_x = C_{(x|y)} D_y w$
 $p_x = C_{(x|y)} p_y$ ④

From ④, we can conclude that $C_{(x|y)} = C_{(y|x)}^T$ ⑤